

■ 3.1.4 Numbers in Other Bases

b^{nnnn}	a number in base b
<code>BaseForm[x, b]</code>	print with x in base b

Numbers in other bases.

When the base is larger than 10, extra digits are represented by letters a–z.

The number 100101_2 in base 2 is 37 in base 10.

```
In[1]:= 2^^100101
Out[1]= 37
```

This prints 37 in base 2.

```
In[2]:= BaseForm[37, 2]
Out[2]= 100101
           2
```

Here is a number in base 16.

```
In[3]:= 16^^ffffaa00
Out[3]= 4294945280
```

You can do computations with numbers in base 16. Here the results are given in base 10.

```
In[4]:= 16^^ffffaa2 + 16^^ff - 1
Out[4]= 16776096
```

This gives the results in base 16.

```
In[5]:= BaseForm[%, 16]
Out[5]= fffb0
           16
```

You can give approximate real numbers, as well as integers, in other bases.

```
In[6]:= 2^^101.100101
Out[6]= 5.578125
```

Here are the first few digits of $\sqrt{2}$ in octal.

```
In[7]:= BaseForm[N[Sqrt[2], 30], 8]
Out[7]= 1.32404746317716746220426276612
           8
```